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24 March 1980

DETAIL SPECIFICATION

SENSOR GROUP, ULTRASONIC MOTION: TRANSCEIVER

RT-1161/FSS-9(V) AND PROCESSOR MX-9444/FSS-9(V)

Inactive for new design after 17 July 1998.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. Scope. This specification covers an ultrasonic motion signal transceiver and an ultrasonic motion signal processor which combine to form an ultrasonic motion sensor, designed to detect the motion of an intruder inside a protected space.

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1. Standards and handbooks. The following standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army CECOM RD&E Center, ATTN: AMSEL-LC-LEO-E-EB, 10115 Gridley Road, Suite 101, Fort Belvoir, VA 22060-5856 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6350
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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-889	-	Dissimilar Metals.

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454	-	General Guidelines for Electronic Equipment.
MIL-HDBK-781	-	Reliability Test Methods, Plans and Environments for Engineering Development, Qualification and Production.

(Unless otherwise indicated, copies of the above standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.)

2.2.2. Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13220E3201	-	Processor, Ultrasonic Motion Signal, MX-9444/FSS-9(V).
TA13220E3211	-	Circuit Card Assembly, Processor.
TA13220E3226	-	Transceiver, Ultrasonic Motion Signal, RT-1161/FSS-9(V).
TA13220E3233	-	Circuit Card Assembly, Transceiver.
TA13220E3247	-	Transducer Kit - Transmit, (White Dot).
TA13220E3248	-	Transducer Kit - Receive, (Blue Dot).
13220E2976	-	Transducer.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: USA CECOM, IEWS Directorate, ATTN: AMSEL-LC-IEW-E-CE, 10115 Gridley Road Suite 208, Fort Belvoir, VA 22060-5818.)

2.3. Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ASQC-Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes.
ISO 10012-1	-	Quality Assurance Requirements For Measuring Equipment. Part 1: Metrological Confirmation System For Measuring Equipment.

(Application for copies should be addressed to: The American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036).

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AMERICAN WELDING SOCIETY (AWS)

D9.1 - Welding of Sheet Metal.

(Application for copies should be addressed to: American Welding Society, 550 NW LeJeune Road, PO Box 351040, Miami, FL 33135.)

2.4. Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. Description. The ultrasonic motion sensor (UMS) shall consist of one processor, ultrasonic motion signal, MX-9444/FSS-9(V) (hereinafter called "processor"), combined with a minimum of one to a maximum of twenty transceivers, ultrasonic motion signal, RT-1161/FSS-9(V) (hereinafter called "transceivers"). The processor shall be in accordance with TA13220E3201 and as specified herein. The transceiver shall be in accordance with TA13220E3226 and as specified herein.

3.1.1. Major components and transducer kits. The major components of and transducer kits for the UMS shall be as follows:

a. Processor, ultrasonic motion signal TA13220E3201.
b. Processor circuit card assembly (CCA) TA13220E3211.
c. Transceiver, ultrasonic motion signal TA13220E3226.
d. Transceiver circuit card assembly (CCA) TA13220E3233.
e. Transducer, drawing 13220E2976.
f. Transducer kit - transmit (white dot) TA13220E3247.
g. Transducer kit - receive (blue dot) TA13220E3248.

When specified (see 6.2), any of the major components or transducer kits listed above shall be furnished separately.

3.1.2. Function. The UMS shall detect movement within interior rooms by processing the Doppler frequency shifts between a transmitted and received ultrasonic signal. This frequency shift is caused by moving objects, within the volume influenced by the sensor. An alarm is generated when the criteria for detection are met. A UMS consisting of one processor and one transceiver, mounted in one corner of an unobstructed 30-foot by 20-foot rectangular room having a ceiling height of 8 to 12 feet, shall generate the alarm signal when movement meeting the criteria for alarm occurs in that room. Up to 20 transceivers can be used with the processor within a volume that is larger or contains obstructions.

3.1.3. Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the UMS. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2. First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

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3.3. Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1. Material deterioration prevention and control. The UMS shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the UMS may be exposed.

3.3.2. Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3. Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs. Used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the UMS.

3.4. Environmental conditions. The UMS and major components or transducer kits when furnished separately shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.3), performance degradation, or malfunction when tested as specified in 4.4.2.5.

3.4.1. Operational.

a. Temperature range of -20 to +150 °F.
b. Humidity up to 95 percent relative.

3.4.2. Non-operational condition and storage.

a. Temperature range of -30 to +165 °F.
b. Humidity up to 95 percent relative.
c. Shock: bench handling, equipment not operating.

3.5. Detailed technical requirements. The UMS and major components or transducer kits when furnished separately shall perform as specified herein under any of the operating test conditions specified herein.

3.5.1. Terminal board connections.

3.5.1.1. Transceiver terminal board. The transceiver terminal board shall be marked with identifying terminal numbers as follows:

Terminal Board 1 (TB1)	Function
1	Transmitter input signal
2	Transmitter input signal return
3	Receiver output signal
4	Receiver output signal return
5	Tamper signal
6	Tamper signal
7	Spare
8	Spare
9	Enclosure ground
10	Spare

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3.5.1.2. Processor terminal boards. The processor terminal boards shall be marked with identifying terminal numbers as follows:

Terminal Board 1 (TB1)	Function
1	Alarm output
2	Alarm output return
3	Spare
4	Spare
5	Tamper signal
6	Tamper signal
7	+20 volts direct current (Vdc) in
8	20 Vdc return
9	Enclosure ground
10	Spare

On TB1, removable jumpers shall be provided across terminals 2 and 3, 3 and 4, and 4 and 5.

Terminal Board 2 (TB2)	Function
1	Transmitter output signal
2	Transmitter output signal return
3	Receiver input signal
4	Receiver input signal return
5	Spare
6	Spare
7	Synchronization (sync) signal output
8	Signal ground
9	Enclosure ground
10	Spare

3.5.2. Functional characteristics.

3.5.2.1. Power. Power for a UMS shall be supplied to the processor from a 20 Vdc regulated power supply. The processor shall supply operating power to the transceiver. Reverse polarity connections to the processor shall not damage the UMS. A UMS consisting of a processor and from one to 20 transceivers shall operate as specified herein on regulated dc voltage between 18 and 22 Vdc with a current drain not exceeding 110 milliamperes (mA).

3.5.2.1.1. Transmitter sound level output. For a UMS consisting of a processor and a single transceiver, the on-axis sound pressure level at a distance of one foot from the transmit transducer shall not be less than 100 decibels (dB) above reference sound pressure level (RSPL) of 0.0002 microbars. The frequency of the signal shall be 26.300 ±0.005 kilohertz (kHz).

3.5.2.2. Performance. The UMS shall be sensitive to broadband ultrasonic noise with frequency components within the passband of the UMS (25.3 to 27.3 kHz) that exceed the RSPL at the receive transducer by 65 dB. The UMS shall be sensitive to the range of Doppler frequencies produced by unidirectional motion of targets with radial velocities between 0.4 feet per second (fps) and 20 fps for specified minimum durations. The specified minimum durations are 1 second at a velocity of 0.4 fps decreasing exponentially to 0.3 second at a velocity of 10 fps. For all velocities from 10 to 20 fps, the specified minimum durations are 0.3 second. The UMS shall perform as specified in 3.5.2.2.1 through 3.5.2.2.4 under the conditions listed below:

a. Number of transceivers connected to the processor: up to 20.
b. Cable between the processor and each transceiver: up to 500 feet of cable, RG-58/U or equivalent.

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3.5.2.2.1. Alarm criteria for noise. The UMS shall be in the alarm state after broadband ultrasonic noise, as described in 3.5.2.2.1.1, that exceeds the RSPL at the receive transducer by 65 dB has been applied to one transceiver in a motion-free area for 5 seconds or more with its range control set to maximum and all other transceivers placed in an acoustically quiet, motion-free area.

3.5.2.2.1.1. Spectrum of broadband ultrasonic noise. The broadband ultra-sonic noise shall be that produced by a band-limited white noise generator producing a uniform noise output in the 25.3-27.3 kHz band when applied to a suitable transducer (see Appendix A, figure 1). The 3 dB bandwidth shall be 5 to 30 kHz with a 6 dB/octave slope outside this bandwidth.

3.5.2.2.2. Alarm criteria for unidirectional motion. For a constant setting of the transceiver range control, the UMS shall be in the alarm state in accordance with table I upon receipt of the ultrasonic Doppler signals simulating the unidirectional motions listed in table I with velocities in the range 0.55 to 17.6 fps. The UMS shall detect such motions in the presence of the ultrasonic clutter signal in accordance with table I. The UMS shall be in the non-alarm state when only the clutter signal is present and when neither the clutter signal nor the unidirectional Doppler signal is present. Unless otherwise specified herein, the performance described in this paragraph shall be obtained with only a single transceiver operating with a processor.

TABLE I. Schedule of UMS states for sums of simulated Doppler signals and clutter signals.

Simulated ultrasonic Doppler. Signal 1/				
Type of motion	Target radial velocity fps 2/	Target duration or period 3/	Ultrasonic clutter signal 4/	UMS state
Unidirectional	0.55	960 ms	None	Alarm
Unidirectional	1.1	780 ms	Present	Alarm
Unidirectional	2.2	570 ms	Present	Alarm
Unidirectional	4.4	380 ms	Present	Alarm
Unidirectional	8.8	310 ms	Present	Alarm
Unidirectional	17.6	300 ms	Present	Alarm
None	-	-	None	Non-alarm
None	-	-	None	Non-alarm
Oscillating	0.55	2 sec	None	Non-alarm
Oscillating	1.1	2 sec	Present	Non-alarm
Unidirectional	1.138	1 sec	None	Non-alarm
Unidirectional	35.2	1 sec	None	Non-alarm

Notes:
1/ The simulated ultrasonic Doppler signal shall be obtained from a Doppler signal synthesizer (see 6.4.4) and shall simulate the listed target motion. The simulated Doppler signal shall exceed the RSPL at the receive transducer by no more than 40 dB, unless otherwise noted.
2/ For approaching and receding radial velocities of 17.6 fps, the simulated ultrasonic Doppler and clutter signals shall be applied to one transceiver, with 19 other transceivers located in an acoustically quiet area connected to the processor by up to 500 feet of RG-58/U cable. The sound pressure level of the simulated ultrasonic Doppler signal at the receive transducer shall not exceed 55 dB above the RSPL. The 19 transceivers and cable may be simulated by suitable electronic circuits having impedances and noise characteristics equivalent to those experienced in a typical string of 19 transceivers.

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3/ The simulated ultrasonic Doppler signal for oscillatory motion shall be obtained from the Doppler signal synthesizer. The signal simulating oscillatory motion shall be:
(a) Continuous.
(b) Cyclic of period 2 seconds (sec).
(c) Zero at its start and symmetric about the zero reference.

4/ The ultrasonic clutter signal shall consist of broadband ultrasonic noise as specified in 3.5.2.2.1.1. The clutter signal shall be continuous and shall exceed RSPL at the receive transducer by at least 45 dB.

3.5.2.2.3. Oscillatory motion. For the minimum setting of the transceiver range control at which the UMS conforms to 3.5.2.2.2, the UMS shall be in the non-alarm state upon receipt of the ultrasonic Doppler signals simulating the oscillatory motions listed in table I. The UMS shall be in the non-alarm state when simulated oscillatory motion appears in the presence of the ultrasonic clutter signal as listed in table I.

3.5.2.2.4. Out-of-band target signals. For the minimum setting of the transceiver range control at which the UMS conforms to 3.5.2.2.2, the UMS shall be in the non-alarm state upon receipt of the ultrasonic Doppler signals simulating the unidirectional motions listed in table I with velocities less than 0.4 fps and more than 20 fps.

3.5.2.3. Detailed circuit requirements.

3.5.2.3.1. Transceiver range control. The transceiver range control shall provide an adjustment of at least 20 dB.

3.5.2.3.2. Alarm duration. When the alarm criteria are met, the UMS shall be in the alarm state for not less than 0.5 seconds nor more than 1.5 seconds after the signals or noise meeting the alarm criteria are removed.

3.5.2.3.3. Alarm output. The resistance across the processor alarm output terminals (TB1-1 and TB1-2) shall be as follows:

a. Non-alarm state: less than 150 ohms.
b. Alarm state: more than 100 kilohms (k ohms).

The alarm output terminals shall be capable of accepting up to 10 mA dc in the non-alarm state and up to 30 Vdc in the alarm state.

3.5.2.3.4. Power interruption. When power to the UMS is interrupted, or the ground wire is disconnected, the processor shall be in the alarm state.

3.5.2.4. Tamper protection. The resistance between terminals 5 and 6 of the transceiver and processor TB1 shall represent a tamper alarm condition by exceeding 100k ohms, or a non-tamper condition by being less than 10 ohms. The tamper alarm condition shall be presented whenever the cover is opened from .062 to .250 inch (flanges engaged) and shall continue until the cover is closed or the tamper switch actuator rod is pulled out to detented position (maintenance position). The tamper switch shall reset by closing the cover.

3.5.2.5. Transceiver and transceiver CCA. The transceiver and the transceiver CCA when furnished separately shall be in accordance with TA13220E3226 and TA13220E3233 respectively and have the following characteristics:

a. Pre-amp gain. With the receive transducer disconnected and a sinusoidal input signal of 26.3 ±0.1 kHz, 10.0 ±0.1 millivolt (mV) peak-to-peak (P-P) applied across the receive jacks, the output voltage at TB1-3 referenced to ground shall be 40 ±15 mV P-P with

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the range control set to position 10. When the range control is set to position 1, the output voltage shall not be greater than 10 ±1 percent of the voltage measured with the range control set to position 10.

b. Terminal board connections to jacks. Continuity (less than 10 ohm resistance) shall exist between terminal board terminals and jacks on the transceiver CCA as follows:

FROM	TO
TB1-1	Transmit jack marked "RED"
TB1-2	Transmit jack marked "BLUE"
TB1-4	Receive jack marked "BLUE"
TB1-9	Chassis ground jack marked "BLACK".

c. Tamper switch. A resistance of 10 ohms or less shall exist between TB1-5 and TB1-6 when the tamper switch plunger is fully depressed or fully extended. A resistance between TB1-5 and TB1-6 of not less than 100k ohms shall exist after the tamper switch plunger is momentarily depressed and then released. The transceiver tamper switch shall meet the requirements specified in 3.5.2.4. When the transceiver CCA is mounted in the transceiver enclosure, the tamper switch shall operate as specified in 3.5.2.4.

d. Transmit and receive transducers . When disconnected from TB1 the transmit and receive transducers in the transceiver shall operate as specified in 3.5.2.7.

3.5.2.6. Processor and processor CCA. The processor and the processor CCA when furnished separately shall be in accordance with TA13220E3201 and TA13220E3211 respectively and shall meet the requirements specified herein. The circuit operation shall be as follows when tested as specified in test method 7 of Appendix A, with 20 ±0.1 Vdc applied to TB1-7, referenced to TB1-8:

a. Operating voltage and current criteria. With the signal generator (see Figure 3) turned off, the following conditions shall exist:

(1) Input current to TB1-7 shall be 37 ±6 milliamperes (mA) DC.
(2) Voltage at the plus side of capacitor C35 referenced to TB1-8 shall be 12 ±0.6 Vdc.
(3) Voltage between the "reference" test point and TB1-8 shall be 6.8 ±0.6 Vdc.
(4) Voltage between the "balance" and "reference" test points shall be 0 ±0.2 Vdc.
(5) If the polarity of the voltage across TB1-7 and TB1-8 is reversed, the input current to TB1-8 shall not exceed 1 milliampere.

b. Alarm criteria for Doppler signals.

(1) The AC signal between TB2-1 and signal ground (TB2-4) shall be 5.7 ±0.2 Vrms at 26.300 ±0.005 kHz.
(2) When the signal generator is turned on and set to provide a voltage, as measured at E2 referenced to TB2-4 (see figure 3), equal to 33 percent ±2 percent of that measured in (1) above, at a frequency 400 ±20 Hz greater than that measured in (1) above, a signal having a frequency (Doppler frequency) of 400 ±20 Hz shall exist at the "noise" test point.
(3) When the attenuators are set to 42 dB, the non-alarm state shall exist.
(4) When the attenuator setting is gradually reduced in 1 dB steps, the alarm state shall occur at an attenuation of 37 ±3 dB.

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c. Alarm criteria for noise. With the signal generator set as in 3.5.2.6b.(2), when the Doppler channel is deactivated by connecting a clip lead from the positive side of C43 to the junction of R52 and R53, and the attenuators are set to 32 ±1 dB, the non-alarm state shall exist. When the attenuation is gradually reduced, the alarm state shall occur at an attenuation of 25 ±3 dB.

d. Log amplifier gain criteria. With the signal generator set as in 3.5.2.6b.(2), and with the Doppler channel deactivation clip lead removed and the attenuators set to obtain 1.0 ±0.1 Vrms at the signal (SIG) test point, a reduction in attenuation of 20 dB shall result in a voltage of 1.8 ±0.3 Vrms at the signal test point.

e. Linear amplifier gain and Doppler filter criteria. With the signal generator set as in 3.5.2.6b.(2) and the attenuators set to 20 dB, a voltage of 2.8 ±0.3 Vrms shall exist between the noise test point and TB2-4. When the attenuation is increased to 40 dB, the voltage at the noise test point shall be reduced by a factor of 10 (±10 percent). When the signal generator frequency is adjusted to obtain a frequency at the noise test point of +800 ±20 Hz, -800 ±20 Hz, +12 ±2 Hz and -12 ±2Hz, the voltage between the noise test point and TB1-4 shall be at least 3 dB less than that when the frequency was 400 ±20 Hz.

f. Gating waveform criteria. With the signal generator set as in 3.5.2.6b.(2) and the attenuators set to 40 dB, pulses of +12 ±1 volts amplitude and 120 ±55 microseconds in duration shall exist at the junction of resistors R51 and R57. The pulses shall occur at the negative peaks of the waveform at the signal test point. Adjusting the signal generator frequency to 400 ±20 Hz below that measured in 3.5.2.6.b(1) shall cause the pulses to occur at the positive peaks of the waveform at the signal test point.

g. Transmitter failure alarm. With the signal generator set as in 3.5.2.6b.(2), when the attenuators are set to 70 dB, the non-alarm state shall exist. An alarm state shall exist when the crystal oscillator assembly red lead is disconnected for 250 milliseconds and greater. Reconnecting the red lead shall restore the non-alarm state within 1.5 seconds.

h. Synchronization output. With the signal generator set as in 3.5.2.6b.(2) and the attenuators set to 70 dB, a symmetrical triangular waveform of 3.0 ±0.3 volts peak to peak shall exist between TB2-7 and TB2-4.

i. Tamper switch. Tamper switch operation shall be identical to that specified in 3.5.2.5c for transceiver tamper switch. The processor shall also meet the requirements specified in 3.5.2.4.

j. Alarm duration. When the alarm criteria are met, the processor shall be in the alarm state for not less than 0.5 seconds nor more than 1.5 seconds after the alarm criteria are no longer met.

3.5.2.7. Transducers. The transducers (white dot and blue dot) shall be in accordance with drawing 13220E2976 and have the following electrical characteristics:

a. White dot (13220E2976-1).

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Type: transmitter.
Minimum beam width at 3 dB points: 40 degrees.
Excitation frequency: 26.30 kHz.
Excitation waveform: square wave, 20 volts peak to peak.
Minimum acoustic output: 26 dB relative to 1 microbar at 1 foot on axis.
Spurious emissions: shall emit no audible energy.

b. Blue Dot (13220E2976-3).

Type: receiver.
Frequency of maximum acoustic-to-electrical (voltage) conversion, loaded with 6.8K ohm resistor: 26.30 ±0.30 kHz.
Acoustic bandwidth at -3 dB points loaded with 6.8k ohm resistor: 1.2 kHz minimum.
Sensitivity: 58 dB relative to 1 volt per microbar.

3.5.2.8. Transducer kit-transmit (white dot). The transducer kit-transmit (white dot) shall be in accordance with TA13220E3247. The transducer shall meet the requirements of 3.5.2.7a.

3.5.2.9. Transducer kit-receive (blue dot). The transducer kit-receive (blue dot) shall be in accordance with TA13220E3248. The transducer shall meet the requirements of 3.5.2.7b.

3.6. Electromagnetic susceptibility. The electromagnetic susceptibility characteristics of the UMS shall conform to MIL-STD-461, methods RS103 and CS101, as modified in 3.6.1 and 3.6.2.

3.6.1. Radiated. Radiated susceptibility shall be conducted in an electric field of 3 volts per meter (V/m) for both steady-state and keyed conditions of the EMI field generating equipment. This field applies to frequencies up to 500 megahertz (MHz). Above 500 MHz, the electric field shall be 1 V/m. Maximum applied frequency shall be 10 Gigahertz.

3.6.2. Conducted. Conducted interference signals shall be applied to both power and signal lines and shall be continuous wave, 30 Hz to 50 kHz.

3.7. Reliability. When tested as specified in 4.4.2.7, the UMS shall have a design mean-time-between-failure (MTBF) of 10,000 hours or greater.

3.8. Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustments, and without selection for fit or performance.

3.9. Identification marking. The UMS and major components and transducer kits shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.10. Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.7) for manufacturing and testing UMSs and major components when furnished separately:

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Item No.	Description	Identification	Quantity
1	Stable base drawings of processor printed wiring board (PWB)	13220E3220, sheets 2, 3, and 4	1 per sheet (see note 1)
2	Stable base drawings of transceiver (PWB)	13220E3240, sheets 2 and 3	1 per sheet (see note 2)

Notes:
1. Provided only when complete UMSs, processors or processor CCAs are procured.
2. Provided only when complete UMSs, transceivers or transceiver CCAs are procured.

3.11. Workmanship. Workmanship shall be in accordance with the guidelines and guidance paragraphs of MIL-HDBK-454, guideline 9.

3.11.1. Welding. Sheet metal welding shall be in accordance with AWS D9.1.

4. VERIFICATION

4.1. Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2. First article inspection.

4.2.1. Examination. Each first article UMS and major components or transducer kits when furnished separately shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.2.2. Tests.

4.2.2.1. Complete UMS. The complete UMS shall be subjected to the tests marked "X" in column 1 of table II. Individual testing of major components of the complete UMS is not required. Failure of any test shall be cause for rejection. Unless otherwise specified (see 6.2), five UMSs shall be subjected to environmental tests (see 4.4.2.5), twelve UMSs shall be subjected to the reliability test (see 4.4.2.7) and, when required (see 6.2), one UMS, selected from the environmental test units, shall be subjected to the electromagnetic susceptibility test (see 4.4.2.6).

4.2.2.2. Major components or transducer kits furnished separately. Major components or transducer kits furnished separately shall be subjected to the tests marked "X" in column 1 of tables III through V. Failure of any test shall be cause for rejection. Unless otherwise specified (see 6.2), five units shall be subjected to each test.

4.3. Conformance inspection.

4.3.1. Sampling. Sampling for tests shall be in accordance with ANSI/ASQC-Z1.4. Sample size shall be determined by using tables I and IIA, special inspection level S-2. A lot shall be accepted when no defects are found and rejected when one or more defects are found. Samples shall be randomly selected from production lots.

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4.3.2. Examination. Each UMS and each major component or transducer kit furnished separately shall be examined for the major defects specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.3. Tests.

4.3.3.1. Individual.

4.3.3.1.1. Complete UMS. Each complete UMS shall be subjected to the tests marked "X" in column 2 of table II. Individual testing of major components of the complete UMS is not required. Failure of any test shall be cause for rejection.

4.3.3.1.2. Major components or transducer kits furnished separately. Each major component or transducer kit when furnished separately shall be subjected to the tests marked "X" in column 2 of tables III through V, as applicable. Failure of any test shall be cause for rejection.

4.3.3.2. Samples.

4.3.3.2.1. Complete UMS. Each sample UMS, selected in accordance with 4.3.1, shall be subjected to the tests marked "X" in column 3 of table II. Individual testing of major components of the complete UMS is not required. Unless otherwise specified, a sample UMS shall consist of one processor and one transceiver. Failure of any test shall be cause for rejection.

4.3.3.2.2. Major components or transducer kits furnished separately. Each sample major component or transducer kit when furnished separately, selected in accordance with 4.3.1, shall be subjected to the tests marked "X" in column 3 of tables III through V, as applicable. Failure of any test shall be cause for rejection.

4.4. Inspection procedure.

4.4.1. Examination. The UMS and major components or transducer kits when furnished separately shall be examined as specified herein for the following major defects:

Major	
101.	Components missing or not as specified (see 3.1, 3.1.1, and 3.1.3).
102.	Assembly not correct (see 3.1, 3.1.1, and 3.1.3).
103.	Materials not as specified (see 3.3).
104.	Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
105.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
106.	Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the UMS or major components or transducer kits when furnished separately (see 3.3.3).
107.	Interchangeability not as specified (see 3.8).
108.	Identification marking missing, incomplete or illegible (see 3.9).
109.	Workmanship not as specified (see 3.11).
110.	Welding not as specified (see 3.11.1).

4.4.2. Tests.

4.4.2.1. Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. Unless otherwise specified herein, the UMS and major components or transducer

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kits when furnished separately shall operate as specified herein without maintenance.

4.4.2.2. Test schedule.

TABLE II. Test schedule for complete UMS.

First article	Quality conformance		Test	Requirement Paragraph	Test Paragraph
	Individual	Sample			
1	2	3	4	5	6
			Functional:		
X	X	–	(a) Power	3.5.2.1	4.4.2.4.1
X	X	–	(b) Performance	3.5.2.2	4.4.2.4.2
X	–	X	(c) Detailed circuit	3.5.2.3	4.4.2.4.3
X	–	X	(d) Tamper switch tolerance	3.5.2.4	4.4.2.4.4
X 1/	–	X	Low temperature	3.4.1 (a), 3.4.2 (a)	4.4.2.5.1
X 1/	–	X	High temperature	3.4.1 (a), 3.4.2 (a)	4.4.2.5.2
X 1/	–	X	Humidity	3.4.1 (b), 3.4.2 (b)	4.4.2.5.3
X 1/	–	–	Shock	3.4.2 (c)	4.4.2.5.4
X 1/	–	–	Electromagnetic susceptibility	3.6	4.4.2.6
X 2/	–	–	Reliability	3.7	4.4.2.7

1/ Reliability test units not subjected to this test.
2/ Environmental test units not subjected to this test.

TABLE III. Test schedule for transceiver and transceiver CCA.

First article	Quality Conformance		Test 1/	Requirement Paragraph	Test Paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	–	Functional	3.5.2.5	4.4.2.4.5
X	–	X	Tamper switch tolerance 2/	3.5.2.4	4.4.2.4.4
X	–	X	Low Temperature	3.4.1 (a), 3.4.2 (a)	4.4.2.5.1
X	–	X	High Temperature	3.4.1 (a), 3.4.2 (a)	4.4.2.5.2
X	–	X	Humidity	3.4.1 (b), 3.4.2 (b)	4.4.2.5.3
X	–	–	Shock 2/	3.4.2 (c)	4.4.2.5.4

1/ The transceiver CCA may be placed in a transceiver enclosure or equivalent to simulate normal environment.
2/ Required for transceiver only.

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TABLE IV. Test schedule for processor and processor CCA .

First article	Quality Conformance		Test 1/	Requirement Paragraph	Test Paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	–	Functional	3.5.2.6	4.4.2.4.6
X	–	X	Tamper switch tolerance 2/	3.5.2.4	4.4.2.4.4
X	–	X	Low Temperature	3.4.1 (a), 3.4.2 (a)	4.4.2.5.1
X	–	X	High Temperature	3.4.1 (a), 3.4.2 (a)	4.4.2.5.2
X	–	X	Humidity	3.4.1 (b), 3.4.2 (b)	4.4.2.5.3
X	–	–	Shock 2/	3.4.2 (c)	4.4.2.5.4

1/ The processor CCA may be placed in a processor enclosure or equivalent to simulate normal environment.
2/ Required for processor only.

TABLE V. Test Schedule for transducer kit.

First article	Quality Conformance		Test 1/	Requirement Paragraph	Test Paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	–	Functional	3.5.2, 3.5.2.9	4.4.2.4.7, 4.4.2.4.8
X	–	X	Low Temperature	3.4.1 (a), 3.4.2 (a)	4.4.2.5.1
X	–	X	High Temperature	3.4.1 (a), 3.4.2 (a)	4.4.2.5.2
X	–	X	Humidity	3.4.1 (b), 3.4.2 (b)	4.4.2.5.3

1/ The transducer kit may be placed in a transceiver enclosure or equivalent to simulate normal environment.

4.4.2.3. Test methods. Tests shall be conducted in accordance with the test methods in Appendix A and as specified herein. Calibration of the test equipment shall be in accordance with ISO 10012-1.

4.4.2.4. Functional.

4.4.2.4.1. Power. The UMS shall be tested in accordance with Appendix A Test Method 1. Nonconformance to 3.5.2.1 shall constitute failure of this test.

4.4.2.4.2. Performance. The UMS shall be tested in accordance with Appendix A Test Method 2. Nonconformance to 3.5.2.2 shall constitute failure of this test.

4.4.2.4.3. Detailed circuit. The UMS shall be tested in accordance with Appendix A Test Method 3. Nonconformance to 3.5.2.3 shall constitute failure of this test.

4.4.2.4.4. Tamper switch tolerance test. The UMS and major components when furnished separately shall be tested in accordance with Appendix A Test Method 4. Nonconformance to 3.5.2.4 shall constitute failure of this test.

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4.4.2.4.5. Transceiver and transceiver CCA. The transceiver and transceiver CCA shall be tested in accordance with Appendix A Test Methods 4 and 6. Nonconformance to 3.5.2.4 and 3.5.2.5 shall constitute failure of this test.

4.4.2.4.6. Processor and processor CCA. The processor and processor CCA shall be tested in accordance with Appendix A Test Methods 4 and 7. Nonconformance to 3.5.2.4 and 3.5.2.6 shall constitute failure of this test.

4.4.2.4.7. Transducer kit-transmit (white dot). The transducer kit (white dot) shall be tested in accordance with Appendix A Test Method 8. Nonconformance to 3.5.2.8 shall constitute failure of the test.

4.4.2.4.8. Transducer kit-receive (blue dot). The transducer kit (blue dot) shall be tested in accordance with Appendix A Test Method 8. Nonconformance to 3.5.2.9 shall constitute failure of the test.

4.4.2.5. Environmental. The UMS and major components or transducer kits when furnished separately shall be subjected to the following environmental tests. The functional test required (see B.3.2.1) shall be the test specified in 4.4.2.4 applicable to the item being furnished, except for the complete UMS. The functional test for the complete UMS shall be the operational test, Test Method 5 of Appendix A.

4.4.2.5.1. Low temperature. Low temperature test shall be conducted in accordance with low temperature test procedure of Appendix B (see B.4.1).

4.4.2.5.2. High temperature. High temperature test shall be conducted in accordance with high temperature test procedure of Appendix B (see B.4.2).

4.4.2.5.3. Humidity. Humidity test shall be conducted in accordance with the humidity test of Appendix B (see B.4.3).

4.4.2.5.4. Shock. Shock test (bench handling) shall be conducted in accordance with shock test of Appendix B (see B.4.4).

4.4.2.5.5. Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.4.2.6. Electromagnetic susceptibility. When specified (see 6.2), one UMS, consisting of one processor and one transceiver, shall be subjected to the following electromagnetic susceptibility tests of MIL-STD-461.

4.4.2.6.1. Radiated susceptibility. Conduct Test Method RS103, except that the electric field requirements of RS103 shall be modified and limited as specified in 3.6.1. All leads (sensor, alarm and power) external to the signal processor enclosure shall be exposed to the electrical field. The leads shall not be less than 4 feet in length.

4.4.2.6.2. Conducted susceptibility. Conduct Test Method CS101 (except level shall be reduced 0.4 volts between 24.5 and 27.5 kHz), as modified in 3.6.2.

4.4.2.6.3. Failure criteria. Nonconformance to 3.6 shall constitute failure of this test.

4.4.2.7. Reliability. Using an upper test MTBF (θ_0) of 10,000 hours and lower test MTBF (θ_1) of 5,000 hours, the UMS shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-HDBK-781, test plan VIIID. Continue testing until either an "accept" or "reject" decision is reached. All test incidents will be scored by the Government. A

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failure is defined as any malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below designated levels.
b. Serious damage to the system.
c. Serious personnel safety hazards.

4.4.2.7.1. Test procedure. Unless otherwise specified (see 6.2), twelve complete UMSs (see 6.2), each consisting of one processor and one transceiver, shall be randomly selected by a Government representative. The tests shall be conducted under ambient laboratory conditions, without temperature, humidity, or vibration cycling. Equipment shall be energized at the power input specified in 3.5.2.1. Operation shall be continuous throughout the test. The 5-second, wideband ultrasonic noise signal specified in 2.1g of Appendix A Test Method 2 shall be applied to the UMS not less than nine times during each 24-hour period, but not more often than once per hour. At the end of the reliability test, the UMS shall be subjected to the tests specified in 4.4.2.4.1 and 4.4.2.4.2. The test shall commence only after successful completion of functional tests of each unit. Parts or components of known limited life, such as batteries, shall have a life stipulated prior to test. Failure of these parts occurring after the end of the stipulated period are non-relevant, but any dependent failures caused thereby are relevant. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified, if practicable. This accounting shall be maintained separately from the failure accounting. Redundant elements, such as indicator lamps, shall not be repaired nor counted as failure until the UMS has failed to perform its intended function. Preventive maintenance procedures specified for the equipment during normal operation shall be performed during the reliability test. No additional preventive maintenance is allowed during this test.

5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1. Intended use. The UMS is intended for use as an interior motion detector and is a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. Nomenclature and quantity of item(s) to be furnished (see 3.1.1).

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d. Technical Data Package List, including revision level, to which each item being furnished (by contractor) is to be built.
e. When a first article is required for inspection and approval, the total number of units required, and the number to be subjected to environmental and reliability tests respectively, if other than specified (see 3.2, 4.2.2.1, 4.2.2.2 and 6.3).
f. When Government-loaned property is not required (see 3.10).
g. Packaging requirements (see 5.1).
h. When electromagnetic susceptibility testing is required, the number of units to be tested if other than one.

6.3. First article. When a first article inspection is required, the UMSs, major components or transducer kits should be preproduction models. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4. Definitions. The following definitions apply throughout this specification.

6.4.1. False alarm. An alarm generated by stimuli other than those for which the test item is designed to alarm.

6.4.2. Tamper alarm. Alarm resulting from removal of the sensor assembly cover or pry-off of the sensor assembly from its mounting.

6.4.3. Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the UMS; evidence of delamination or water penetration into integrated circuits, printed circuit boards, or external components; or physical condition resulting in noncompliance of the item with the provisions of this specification.

6.4.4. Doppler signal synthesizer. For a given basic transmitted ultrasonic signal, this equipment produces an ultrasonic signal equal in frequency to the basic frequency plus or minus Doppler frequencies corresponding to the selectable radial target velocities, and adjustable in amplitude to simulated target range. The output signal may, as desired, simulate a unidirectional target, or one oscillating in place, and may be either steady-state or pulsed, with the pulse duration selectable. The output signal may be hard-wired to the load or radiated (air-coupled) to the load by close transducer spacing.

6.4.5. Wideband ultrasonic signal generator. The wideband ultrasonic signal generator (WUSG) produces wideband ultrasonic energy, continuous wave (CW) or pulsed, and adjustable in amplitude. The output signal may be hard-wired to the load or radiated (air-coupled) to the load.

6.4.6. Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the objectives of the product and assure interchangeability of the repair parts.

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6.5. Quality assurance provisions (QAP) records. The contracting officer should include a clause requiring the contractor to maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. Such clause should also require that the records include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable.

6.6. Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.7. Government-loaned property. The contracting officer should arrange to lend the property specified in 3.10 when applicable (see 6.2).

6.8. Subject term (key word) listings.

Intrusion detection
Physical security
Sensor
Motion sensor
Ultrasonic motion sensor

6.9. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR4
Navy – SH
Air Force – 99

Preparing activity:
Army – CR4

Project 6350-0223

Review activities:
Army – ER
Navy – CG, MC
DLA – GS

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APPENDIX A

TEST METHODS

FOR

ULTRASONIC MOTION SENSOR GROUP

A.1. SCOPE

A.1.1. Scope. The test methods contained within this appendix are to determine whether components of the Ultrasonic Motion Sensor Group and major components or transducer kits procured under this specification conform to the requirements specified. Figures 1 through 6 are required to facilitate testing of the group or major components or transducer kits. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2. APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

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TEST METHOD 1

POWER TEST

1. TEST APPARATUS:

a. Power supply, dc - constant voltage, 0 to 25 Vdc, 120 mA minimum current, line and load regulation 0.1 percent or less, maximum ripple and noise 5 mV P-P.
b. Ammeter, dc - 199 mA minimum, resolution 0.1 mA, accuracy 1 percent of reading ±2 LSD (2 times value of least significant digit), maximum voltage burden 0.3 Vdc, or equivalent analog unit.
c. Voltmeter, dc - 3 1/2 digit or greater, resolution 10 mVdc , accuracy 0.5 percent of reading ±2 LSD, or equivalent analog meter. (Note: LSD = the smallest digit (value) that can be read on the range setting being used).

2. TEST PROCEDURE:

a. Place the processor in operation by applying 20 ±0.3 Vdc through the ammeter to terminals TB1-7 (positive) and TB1-8. The current shall not exceed 43 mA.
b. Connect 20 transceivers to TB2 of the processor. The total current shall not exceed 110 mA.
c. Reverse the polarity on terminals TB1-7 and TB1-8 for 10 seconds. The current shall not exceed 1 mA.
d. Reconnect as in step b. The current shall not exceed 110 mA.
e. Measure the voltage between Test Point 7 (TP7) (positive) and TP3 (negative) with a voltmeter. The voltage shall be 12 ±1 Vdc.
f. Vary the voltage across TB1-7 and TB1-8 from 18 to 22 V. The voltage between TP7 and TP3 (12 ±1 Vdc) shall not vary more than ±100 mV.

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TEST METHOD 2

PERFORMANCE TEST

1. TEST APPARATUS:

a. Power supply, dc - constant voltage, continuously variable, 0 to 24 Vdc, 120 mA minimum current capacity, line and load regulation 0.1 percent or less, maximum ripple 5 mV P-P.
b. Power supply, dc - constant voltage, continuously variable, 0 to 30 Vdc, 200 mA minimum current capacity, line and load regulation 0.5 percent or less, maximum ripple 20 mV P-P.
c. Voltmeter, dc - 3 1/2 digit or greater, resolution 10 mVdc (range 19.99 V), minimum accuracy 0.5 percent of reading ±2 LSD.
d. Resistor - 2.7k ohms ±5 percent, 1/2 watt.
e. Wideband ultrasonic signal generator (WUSG) - Signal generator producing ultrasonic broadband noise of variable amplitude, hard-wire or radiated (see 6.4.5 for definition; see figure 1 for an alternative schematic).
f. Sound level meter with microphone - measuring range 35 dB to 140 dB minimum SPL with respect to RSPL (.0002 microbars), frequency weighting A, linear and all pass, frequency response 20 Hz to 50 kHz.
g. Oscilloscope - 20 MHz bandwidth, vertical sensitivity 5mV per div. to 5 V per div., sweep rate < 1 microsecond per div. to > 1 second per div., internal/external trigger, ac and dc coupled inputs.
h. Doppler signal synthesizer - as required to perform test procedure (see also 6.4.4).

2. TEST PROCEDURE:

2.1. Alarm criteria for noise.

a. Place 19 of 20 transceivers in an acoustically quiet, motion-free area. One test transceiver shall be excited by the synthesizer test signals.
b. Connect all 20 transceivers to TB2 of the processor by 500 feet of cable. Place the sound level meter microphone in close proximity to the test transceiver receive transducer.
c. Set all transceiver range controls to position 10.
d. Place the UMS in operation by applying 20 ±0.3 Vdc to processor terminals TB1-7 and TB1-8.
e. Operate the WUSG to radiate broadband ultrasonic noise to the microphone and test transceiver. With the sound level meter passband adjusted for the frequency octave centered at 31.5 kHz, adjust the WUSG output level to read 65 dB above RSPL (0.0002 microbar) on the sound level meter.
f. Turn off the ultrasonic noise and momentarily short circuit capacitor C54 (56 microfarads) to reset the noise detector.
g. Turn on the ultrasonic noise for 5 seconds. The UMS shall alarm.

2.2. Alarm criteria for unidirectional motion.

2.2.1 Target signals and background clutter.

a. With the processor and transceiver interconnected as in 2.1b, air-couple the transceiver transmit transducer to the Doppler signal synthesizer input.
b. Hard-wire the WSUG output to be summed with the Doppler signal

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synthesizer output, prior to the signal synthesizer air-coupled output.

c. Apply 30 (+0, -0.3) Vdc, through 2.7k ohms, to processor terminals TB1-1 and TB1-2, and connect oscilloscope across these terminals to monitor alarm function.
d. With UMS in operation as in 2.1d., air-couple (closely spaced) summed WUSG and synthesizer output signals to sound level meter microphone.
e. With Doppler signal synthesizer turned off, operate WSUG to produce broadband ultrasonic noise. Adjust WUSG output level to obtain a reading of 45 dB SPL on the sound level meter. This broadband noise level shall be used in all subsequent steps involving severe clutter.
f. With the WUSG turned off, operate the Doppler signal synthesizer in the steady-state mode at 0.55 fps. Adjust the signal synthesizer output level to obtain a reading of 40 dB SPL on the sound level meter (worst-case target return). This target return signal level shall be used in all subsequent steps involving target signals.
g. Remove the air-coupled, summed signal-and-clutter output from the sound level meter microphone and apply it to the transceiver receive transducer.
h. Turn off the WUSG; operating the Doppler signal synthesizer in the pulsed mode, apply a 0.55 fps target signal (no clutter) for a duration of 960 ms; the UMS shall alarm. Adjust the transceiver range control to the lowest setting at which an alarm can be obtained.
i. Turn on the WUSG. Pulsing the synthesizer, apply 1.1 fps target signal (severe clutter) for a duration of 780 ms. The UMS shall alarm. If necessary, increase the range control to the setting required to obtain an alarm. Do not decrease the range control setting.
j. Repeat step i. for a 2.2 fps target signal and a duration of 570 ms.
k. Repeat step i. for a 4.4 fps target signal and a duration of 380 ms.
l. Repeat step i. for a 8.8 fps target signal and a duration of 310 ms.
m. Increase Doppler signal synthesizer output level to 55 dB above RSPL. Repeat step i. for a 17.6 fps target signal and a duration of 300 ms. The alarm duration, as indicated on the oscilloscope, shall be from 0.5 to 1.5 seconds.
n. Repeat step m. for -17.6 fps target signal.

2.2.2. Background clutter with no target signals.

a. Having performed 2.2.1, leave the transceiver range control at the position to which it was set at the end of 2.2.1n.
b. With the same test equipment configuration as in 2.2.1n. and the same signal levels as in 2.2.1l., turn off the WUSG. The UMS shall not alarm.

2.2.3. Oscillating targets.

a. Leave the transceiver range control as in 2.2.2 a.
b. With the same test equipment configuration and signal levels as in 2.2.1, turn off the WUSG.
c. Operate the Doppler signal synthesizer in the steady state oscillating mode, at 0.5 Hz, for a target velocity of 0.55 fps (no clutter). The UMS shall not alarm.

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d. Turn on the WUSG. Operate the signal synthesizer in steady state oscillating mode at 0.5 Hz, for a target velocity of 1.1 fps (severe clutter). The UMS shall not alarm.

2.2.4. Out-of-band target signals.

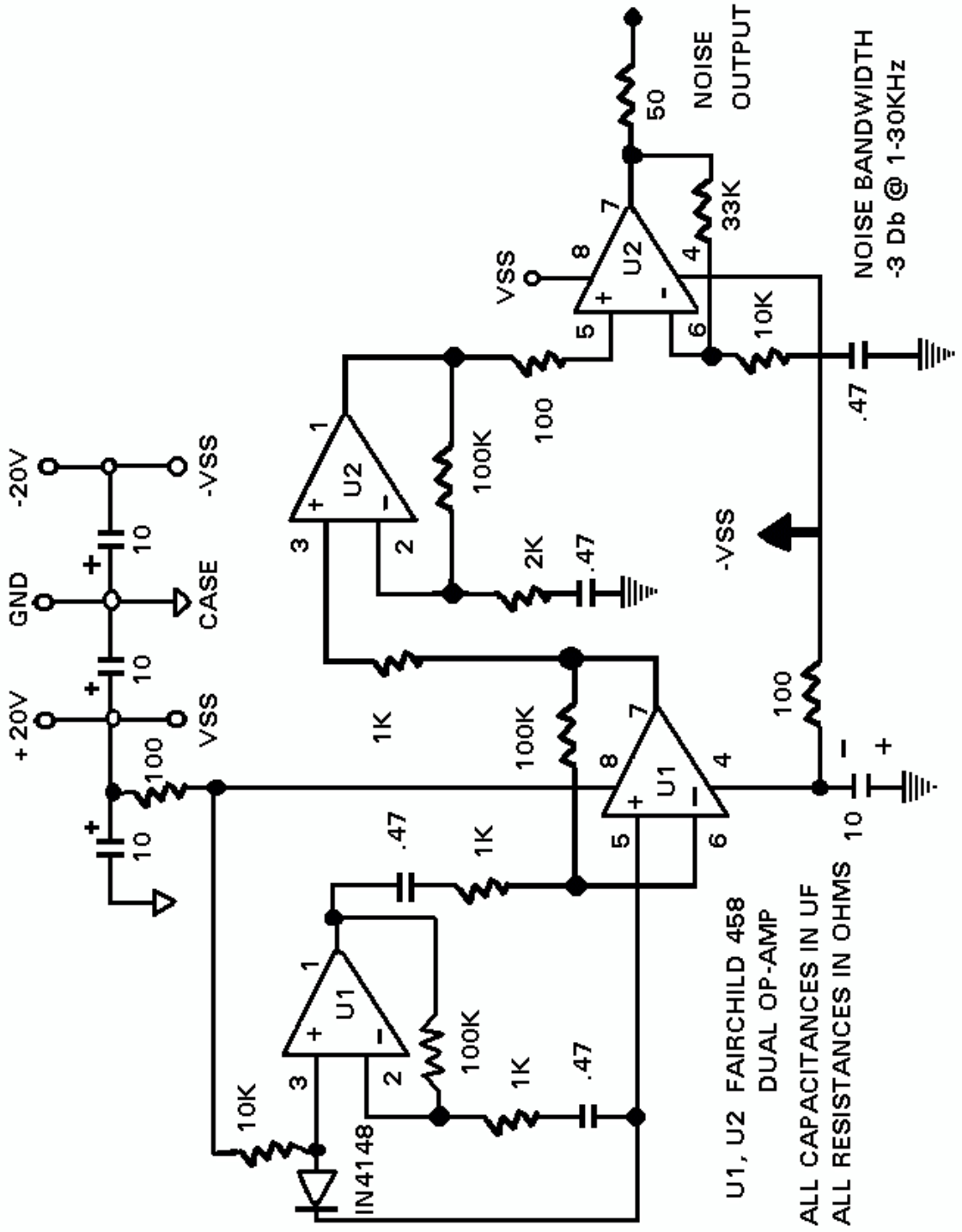
a. Leave the transceiver range control as set in 2.2.2a.
b. Perform step 2.2.3b.
c. Operate the signal synthesizer in the pulsed mode to apply a 0.138 fps target signal (no clutter) for a duration of 1 second. The UMS shall not alarm.
d. Repeat step c. with a 35.2 fps target signal. The UMS shall not alarm.

2.2.5. Transmitter power output.

a. With one transceiver connected to a processor, measure transmitter output using the sound level meter. Place meter microphone 1 foot away from transmitter transducer. Sound level shall be equal or greater than 100 dB above RSPL.

Note: To minimize error in meter indication due to ambient noise, the UMS shall incorporate an external narrow band filter of ±5 kHz bandwidth entered at 26.3 kHz, or shall be measured using sound pressure meter with octave bandpass filter centered at 31.5 kHz.

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TEST METHOD 3

DETAILED CIRCUIT TEST

1. TEST APPARATUS:

a. Power supply, dc - constant voltage, continuously variable, 0 to 25 Vdc, 120 mA minimum current capacity, line and load regulation 0.1 percent or less, maximum ripple 5 mV P-P.
b. Power supply, dc - constant voltage, continuously variable, 0 to 30 Vdc, 100 mA minimum current capacity, line and load regulation 0.5 percent or less, maximum ripple 20 mV P-P.
c. Voltmeter, dc - 3 1/2 digit or greater, resolution 10 mVdc (range 19.99 V), minimum accuracy 0.5 percent of reading ±2 LSD.
d. Resistor - 2.7k ohms ±5 percent. 1/2 watt.
e. Oscilloscope - 20 MHz bandwidth, vertical sensitivity 5 mV per div. to 5 V per div., sweep rate from not greater than 1 microsecond per div. to not less than 1 second per div., internal/external trigger, ac and dc coupled inputs.
f. Doppler signal synthesizer - as required to perform test procedure (see also 6.4.4).
g. Voltmeter, ac - Hewlett Packard Model 403B or equivalent.
h. Ammeter, dc - 199 mA minimum, resolution 0.1 mA, accuracy 1 percent of reading ±2 LSD, maximum voltage burden 0.3 Vdc.

2. TEST PROCEDURES:

2.1. Transceiver range control.

a. With processor and transceiver interconnected, air-couple the transceiver transmit transducer to the Doppler signal synthesizer input, and air-couple the signal synthesizer output to the transceiver receive transducer. Set the transceiver range control to maximum (position 10).
b. Monitor the processor noise test point (marked "NOISE" on circuit card) with ac voltmeter.
c. Place the UMS in operation by applying 20 ±0.3 Vdc to terminals TB1-7 and TB1-8.
d. Apply a steady state synthesized 4.4 fps target signal, at a level that produces a Doppler signal of 2.5 Vrms at the "NOISE" test point.
e. Turn the transceiver range control to minimum (position 1). The signal level at the "NOISE" test point shall decrease by at least 20 dB.

2.2. Alarm duration.

a. With equipment connected as in step 2.1 a., apply 30 (+0, -0.3) Vdc, through 2.7k ohms, to processor terminals TB1-1 and TB1-2 and connect oscilloscope across these terminals.
b. With transceiver range control set to Position 7 and UMS in operation, apply a pulsed synthesized 4.4 fps target signal for a duration of 500 ms, at the signal level in 2.1d.
c. The oscilloscope shall indicate an alarm for a duration of 1 ±0.5 seconds.

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2.3. Alarm output.

a. With equipment connected as in 2.1a, apply 30 (+0, -0.3) Vdc, through 2.7k ohms and the ammeter, to processor terminals TBl-l and TB1-2 and connect dc voltmeter across these terminals.
b. Repeat Step 2.2(b).
c. In non-alarm state, ammeter shall indicate at least 10 mA, demonstrating 10 mA capability and resistance less than 150 ohms.
d. Momentarily interrupt 20 Vdc power, causing UMS to alarm. The dc voltmeter shall read 30 (+0, -1.2) Vdc, demonstrating 30 Vdc hold and resistance greater than 100k ohms.

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TEST METHOD 4

TAMPER SWITCH TOLERANCE TEST

1. TEST APPARATUS:

a. Ohmmeter - resolution 0.5 ohms below 10 ohms, accuracy 3 percent.
b. Test covers, with hole for meter lead access, for processor and transceiver.

2. TEST PROCEDURE:

2.1. Processor.

a. Connect an ohmmeter between the processor tamper terminals TB1-5 and TB1-6.
b. Place the test cover on the processor. The resistance across the tamper terminals TB1-5 and TB1-6 shall be less than 10 ohms.
c. Raise the cover of the processor more than .062 inch but less than .250 inch (flanges engaged). The resistance between TB1-5 and TB1-6 shall increase to greater than 100K ohms.
d. The resistance between TB1-5 and TB1-6 shall be greater than 100K ohms when the cover is removed.
e. Pull the tamper switch fully out and lock. The resistance between TB1-5 and TB1-6 shall decrease to less than 10 ohms.

2.2. Transceiver.

a. Connect an ohmmeter to transceiver tamper terminals TB1-5 and TB1-6.
b. Place the test cover on the transceiver. The resistance across the tamper terminals TB1-5 and TB1-6 shall be less than 10 ohms.
c. Raise the cover on the transceiver more than .062 inch but less than .250 inch (flanges engaged). The resistance across TB1-5 and TB1-6 shall increase to greater than 100K ohms.
d. The resistance between TB1-5 and TB1-6 shall be greater than 100K ohms when the cover is removed.
e. Pull the tamper switch fully out and lock. The resistance between TB1-5 and TB1-6 shall decrease to less than 10 ohms.
f. Repeat steps a. through e. for each transceiver.

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TEST METHOD 5

OPERATIONAL TEST

1. TEST APPARATUS:

a. Power supply, dc - constant voltage, continuously variable, 0 to 24 Vdc, 120 mA minimum current capacity, line and load regulation 0.1 percent or less, maximum ripple 5 mV P-P.
b. Power supply, dc - continuously variable, 0 to 30 Vdc, 100 mA minimum current capacity, line and load regulation 0.5 percent or less, maximum ripple 20 mV P-P.
c. Voltmeter, dc - 3 1/2 digit or greater, resolution 10 mVdc (range 19.99 V), minimum accuracy 0.5 percent of reading ±2 LSD.
d. Resistor - 2.7k ohms ±5 percent. 1/2 watt.
e. Capacitor - 0.1 microfarad ±10 percent.
f. Sound level meter with microphone - measuring range 35 dB to 140 dB minimum SPL with respect to RSPL (.0002 microbars), frequency weighting A, C, linear and all pass, frequency response 20 Hz to 50 kHz or more.
g. Oscilloscope - 20 MHz bandwidth, vertical sensitivity 5mV per div. to 5 V per div., sweep rate < 1 microsecond per div. to > 1 second per div., internal/external trigger, ac and dc coupled inputs.
h. Doppler signal synthesizer - as required to perform test procedure (see also 6.4.4).
i. Frequency counter - 100 Hz to 30 kHz.
j. Voltmeter, ac - Hewlett-Packard 403B or equivalent.

2. TEST PROCEDURE:

a. With processor and transceiver interconnected, hard-wire Doppler signal synthesizer input to transceiver terminals TB1-1 and TB1-2 and monitor this connection with frequency counter. Set transceiver range control to position 7.
b. Hard-wire signal synthesizer output, through a 0.1 microfarad capacitor, to the preamplifier input (transceiver red and blue pins). Disconnect receive transducer.
c. Apply 30 (+0, -0.3) Vdc, through 2.7k ohms, to processor terminals TB1-1 and TB1-2, and connect oscilloscope across these terminals.
d. Place UMS in operation by applying 20 ±0.3 Vdc to processor terminals TB1-7 and TB1-8.
e. Measure transmit frequency with counter.
f. Monitor "NOISE" test point with ac voltmeter and operate Doppler signal synthesizer in steady-state mode at 4.4 fps target velocity. Adjust signal synthesizer output level to obtain Doppler signal of 2.5 Vrms at the "NOISE" test point. This synthesizer output level shall be used in steps g. and h.
g. Operate Doppler signal synthesizer in pulsed mode and apply a 4.4 fps signal for a 400 ms duration. The UMS shall alarm. The alarm duration (as indicated on the oscilloscope) shall be between 0.5 and 1.5 seconds.
h. Operate signal synthesizer in steady-state oscillating mode at 0.5 Hz, for a target velocity of 1.1 fps. The UMS shall not alarm.

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TEST METHOD 6

TRANSCEIVER AND TRANSCEIVER CIRCUIT CARD ASSEMBLY

(TA13220E3226 AND TA13220E3233)

1. TEST APPARATUS:

a. Power supply - Hewlett-Packard model 6202B or equivalent.
b. Oscilloscope - Tektronix, type 465 or equivalent.
c. Signal Generator - Wavetek, Model 110 or equivalent.
d. Ohmmeter - Simpson 260 or equivalent.
e. Resistor - 6.8k ±5 percent, 1/8 watt.
f. Capacitor - 0.15 microfarad ± 5 percent.
g. Voltmeter - dc (M1), Scope model DVM-638 or equivalent.
h. Frequency counter - 20 to 30 kHz
i. Test fixture - (see figure 2).

2. TEST PROCEDURE:

2.1. Receiver pre-amp gain.

a. Disconnect the leads to both transducers (when testing transceiver). Arrange the equipment as shown in figure 2. Adjust dc power supply as necessary to obtain a voltage of 7.0 ±0.1 V, as shown on M1.
b. Apply a sinusoidal signal of 26.3 +0.1 kHz, 10.0 ±0.2 mV peak-to-peak, through a 0.15 microfarad dc blocking capacitor to the pre-amp input terminal as shown.

NOTE: a negative dc bias (-0.29 Vdc, variable) will develop at the red input terminal during normal operation. Avoid connecting any "low pass" or low impedance devices to this terminal.

c. With the range control set to position 10, measure the pre-amp output voltage at TB1-3. This voltage shall be 40 ±15 mV peak-to-peak.
d. With the range control set to position 1, measure the pre-amp output voltage at TB1-3. The voltage shall not be greater than 11 percent of the voltage measured in 2.1c.
e. Turn the range control through positions 1 through 10 several times. The output voltage response observed as in steps c. and d. shall be smooth and continuous and any particular setting shall result in a consistently repeatable output voltage.

2.2. Continuity tests.

a. With an ohmmeter, check for continuity (less than 10 ohms) across the following terminals. Note that the identification of the receive and transmit transducer terminals may not be marked on the printed circuit board. Refer to figure 2, which shows the correct location.

FROM	TO
TB1-1	Transmit jack marked "RED".
TB1-2	Transmit jack marked "BLUE".
TB1-4	Receive jack marked "BLUE" (board ground).
TB1-9	Chassis ground jack marked "BLACK".

2.3. Tamper switch.

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a. Connect an ohmmeter across the tamper terminals TB1-5 and TB1-6 of the transceiver CCA.
b. Depress the tamper switch. The resistance across TB1-5 and TB1-6 shall be less than 10 ohms.
c. Release the tamper switch. The resistance across TB1-5 and TB1-6 shall be greater than 100k ohms.
d. Pull the tamper switch fully out and lock. The resistance across TB1-5 and TB1-6 shall be less than 10 ohms.

2.4. Transmit transducer test.

NOTE: This test shall be performed for transceiver only.

a. Perform Test Method 8 for the white dot (transmit) transducer and blue dot (receive) transducer.
b. After completing tests of both transducers, reconnect leads disconnected in step 2.1 a.

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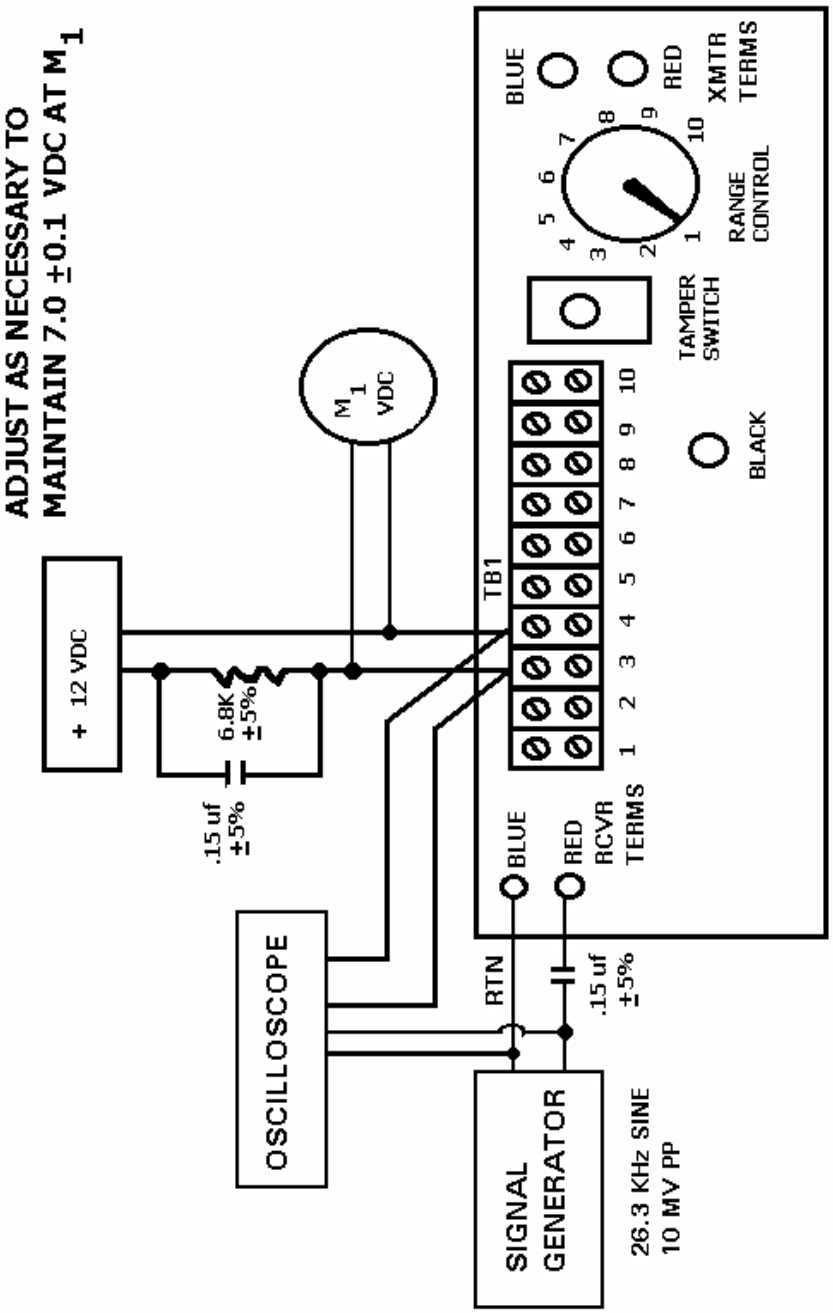


FIGURE 2. Test figure for transceiver and transceiver CCA.

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TEST METHOD 7

PROCESSOR AND PROCESSOR CIRCUIT CARD ASSEMBLY

(TA13220E3201 AND TA13220E3211)

1. Test apparatus.

a. Power supply - Hewlett-Packard model 6216C or equivalent.
b. Signal Generator - Hewlett-Packard model 200CD or equivalent.
c. Oscilloscope - Tektronix model 2235 or equivalent.
d. Electronic counter - Hewlett-Packard model 5300A/5304A or equivalent.
e. Step attenuator - Hewlett-Packard model 355C or equivalent.
f. Step attenuator - Hewlett-Packard model 355D or equivalent.
g. AC voltmeter - Hewlett-Packard Model 403B or equivalent.
h. Multimeter - Fluke Model 8020A or equivalent.
i. Resistor (2 required) - 2k ohms ± 1 percent, 1/8 watt.
j. Resistor - 22k ohms ± 1 percent, 1/8 watt.
k. Resistor - 50 ohms ± 1 percent, 1/8 watt.
l. Capacitor - 1 microfarad ± 10 percent, 50 WVDC.
m. Test fixture - (see figure 3).

2. Test procedure.

2.1. Operating voltage and current criteria.

a. Connect the processor circuit card and test apparatus as shown in figure 3.
b. With signal generator off, turn power supply on and set for +20 ±0.1 Vdc. Measure the input current to TB1-7. It shall be 37 ±6 mA dc.
c. Measure the voltage at the plus side of capacitor C35 referenced to TBl-8. It shall be +12 ±0.6 Vdc.
d. Measure the voltage between the reference test point (marked "REFERENCE" on CCA) and TB1-8. It shall be 6.8 ±0.6 Vdc.
e. Measure the voltage between the balance and reference test points (balance test point marked "BALANCE" on CCA). It shall be 0 ±0.2 Vdc.
f. Disconnect the dc power input leads to TB1-7 and TBl-8, insert ammeter in series with + lead, and reverse the leads (+ lead connected to TB1-8). Measure the reverse current. It shall not exceed one milliampere. Restore to proper configuration.

2.2. Alarm criteria for doppler signals.

a. Measure the ac signal frequency and amplitude at TB2-1 referenced to signal ground (TB2-4). It shall be 26.300 ±0.005 kHz at 5.7 ±0.2 Vrms.
b. Connect the ac voltmeter between the signal generator output (E2) and TB2-4. Turn the signal generator on and set frequency to 400 ±20 Hz greater than that measured in 2.2a. Adjust the signal generator amplitude to a value equal to 33 ±2 percent of that measured in 2.2a.
c. Connect the counter to the "NOISE" test point. The frequency shall be 400 ±20 Hz.
d. Connect the multimeter (ohms scale) to terminals TBl-l and TBl-2. Reduce attenuator setting to 42 dB. The resistance across TB1-1 and TB1-2 shall be 100 ±10 ohms (non-alarm state).

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NOTE: Allow at least 60 seconds for processor to stabilize when large attenuation steps are made.

e. Reduce attenuators in 1 dB steps, allowing several seconds between steps. The resistance shall change to at least 100K ohms (alarm state) at an attenuation not greater than 40 nor less than 34 dB.

2.3. Alarm criteria for noise.

a. Restore equipment configuration and settings to those existing after step 2.2c. The noise for this test will be represented by a single frequency from the signal generator.
b. Connect a clip lead from the positive side of C43 to the junction of R52 and R53 to deactivate the Doppler channel for this test.
c. Set attenuators to 32 ±1 dB. Resistance across TB1-1 and TB1-2 shall be 100 ±10 ohms (non-alarm state).
d. Decrease attenuation in 1 dB steps. Resistance shall become greater than 100k ohms when attenuation is in the range of 25 ±3 dB. (See note after step 2.2d).
e. Remove clip lead previously installed in step 2.3b.

2.4. Log amplifier gain criteria.

a. Configure the equipment as in 2.3a.
b. Connect the ac voltmeter to the processor "SIG" test point and set attenuators to obtain 1.0 ±.1 Vrms at the "SIG" test point.
c. Remove 20 dB of attenuation. Observe voltage at the "SIG" test point. It shall be 1.8 ±.3 Vrms.

2.5. Linear amplifier gain and doppler filter criteria.

a. Configure the equipment as in 2.3a.
b. Connect the ac voltmeter to the "NOISE" test point of the processor, set the attenuators to 20 dB and measure the voltage. The voltmeter shall indicate 2.8 ±.3 Vrms.
c. Increase the attenuation from 20 to 40 dB and measure voltage. The voltage shall be reduced by a factor of 10 (±10 percent).
d. Keeping attenuation and signal amplitude from signal generator constant, vary signal generator frequency and measure ac voltage at "NOISE" test point for (Doppler) frequencies at that test point of +800 ±20 Hz, -800 ±20 Hz, +12 ±2 Hz, and -12 ±2 Hz. The ac voltage at the "NOISE" test point shall be down by at least 3 dB at each of these frequencies.

Note: "+" and "-" signs denote signal generator frequencies greater than and less than respectively the signal measured in 2.1a.

2.6. Gating waveform criteria.

a. Configure the equipment as in 2.3a.
b. Connect channel 1 of the oscilloscope to the junction of R51 and R57 and channel 2 to the "SIG" test point.
c. Channel 1 shall indicate a pulse amplitude of 12 ±1 volts. Each pulse duration shall be 120 ±55 microseconds.
d. Switch on channel 2 of the oscilloscope (dual trace or chopped mode operation). The pulses observed on channel 1 shall occur at the negative peaks of the sine wave observed on channel 2 when the signal generator is set to 400 ±20 Hz greater than that obtained in 2.2a. Change the signal generator frequency to 400 ±20 Hz less than that measured in 2.2a. The pulses observed on channel 1

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shall occur at the positive peaks of the sine wave observed on channel 2.

2.7. Transmitter failure alarm test.

a. Configure the equipment as in 2.3a.
b. Set the attenuators to 70 dB. A resistance of 100 ±10 ohms shall exist between TB1-1 and TB1-2 (non-alarm state).
c. The processor shall generate an alarm condition (at least 100 k ohms between TB1-1 and TB1-2) when the frequency control transducer red lead is disconnected for any duration above 250 milliseconds.
d. Restoration of the frequency control transducer connection shall result in a processor non-alarm condition (100 ohm ±10 ohm between TB1-1 and TB1-2) within 1.5 seconds.

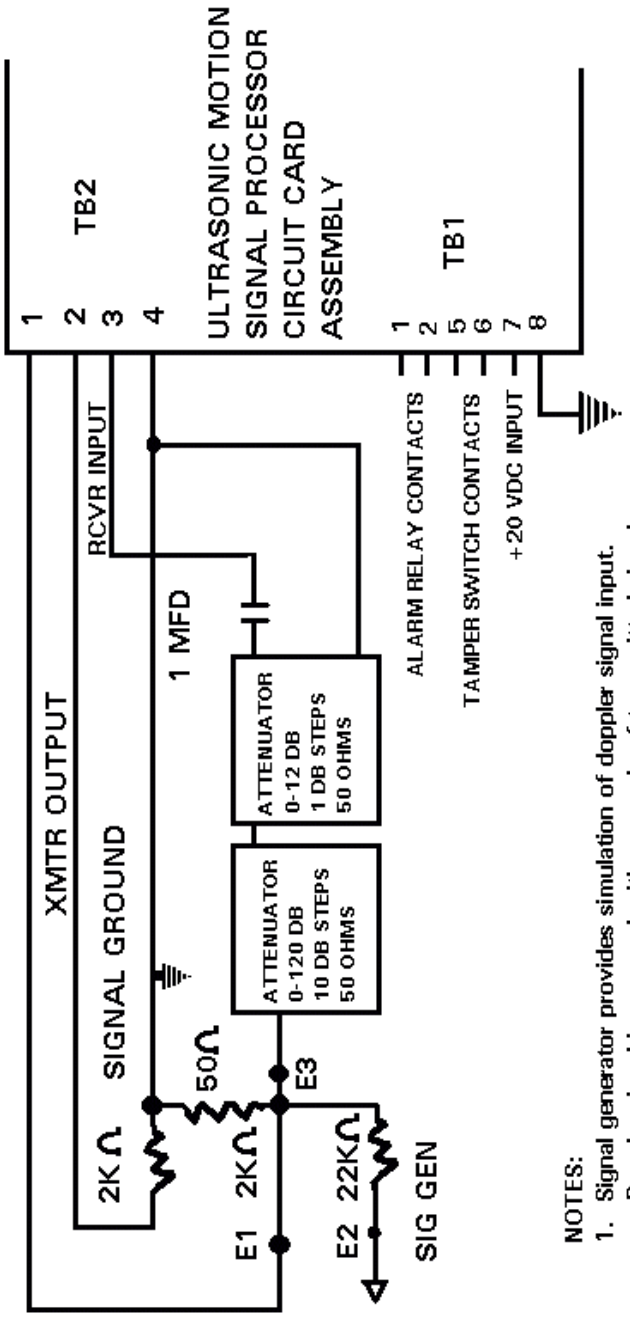
2.8. Synchronization output test.

a. Configure the equipment as in 2.3a.
b. Connect channel 1 of the oscilloscope to TB2-7.
c. A symmetrical triangular wave form of 3.0 ±0.3 volts peak to peak shall be observed.

2.9. Tamper switch test.

a. Depress the tamper switch plunger. A resistance of 10 ohms or less shall occur between TB1-5 and TB1-6.
b. Pull the tamper switch plunger to the full out position. A resistance of 10 ohms or less shall occur between TB1-5 and TB1-6.
c. Depress and release the tamper switch plunger. A resistance of at least 100k ohms shall exist between TB1-5 and TB1-6.

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NOTES:

1. Signal generator provides simulation of doppler signal input. Doppler signal is summed with a sample of transmitted signal to provide a clutter plue doppler input to receiver.
2. Alarm condition shall exist when alarm relay contacts are open (>100K ohms). No alarm condition shall exist when alarm relay contacts are closed (100 ± 10 ohms).
3. Tamper condition shall exist when tamper switch contacts are open (>100K ohms). Non-tamper condition shall exist when tamper switch contains are closed (<10 ohms).

FIGURE 3. Test figure for processor and processor CCA.

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TEST METHOD 8

TRANSDUCER

(C1322OE2976-1 and -3)

1. TEST APPARATUS:

a. Power supply - Hewlett-Packard Model 6216C or equivalent.
b. Anechoic chamber - 3 ft. by 3 ft. by 3 ft.
c. Measuring amplifier - B&K type 2606 w/type 4135 microphone and type 2618 preamplifier or equivalent.
d. Signal generator - Hewlett-Packard, Model 3310A or equivalent.
e. Frequency counter - Hewlett-Packard, Model 5245L or equivalent.
f. Spectrum analyzer - Hewlett-Packard, Model 3580A or equivalent.
g. Attenuators - Hewlett-Packard, Models 355C and 355D or equivalent.
h. Test fixtures - (see figures 4 and 5).

2. TEST PROCEDURE:

2.1. Transducer (13220E2976-1) white dot.

a. Arrange the equipment as shown in figure 4.
b. Calibrate the measuring amplifier in accordance with manufacturer's procedure.
c. Set measuring amplifier to provide an indication of +20 dB.
d. Connect the output from the signal generator to the transducer under test.
e. Minimum acoustic output shall be +20 dB relative to 1 microbar at 1 foot on axis.

2.2. Transducer (13220E2976-3) blue dot.

a. Arrange the equipment as shown in figure 5.
b. Calibrate the measuring amplifier in accordance with manufacturer's procedure.
c. Check calibration of spectrum analyzer.
d. Set spectrum analyzer controls as indicated below:
   (1) Amplitude mode: log 10 dB/div.
   (2) Amplitude reference level: normal.
   (3) Input sensitivity: +30.
   (4) Resolution bandwidth: 100.
   (5) Sweep time/div.: 0.5 sec.
   (6) Sweep mode: repetitive
   (7) Frequency span/div.: 0.
   (8) Frequency: 26.3 kHz.
e. Adjust the tracking oscillator level control of the spectrum analyzer to indicate 75 dB SPL on the measuring amplifier scale.
f. Set the measuring amplifier input attenuator to 10 volts. Adjust the measuring amplifier gain control until the line on the spectrum analyzer is at -58 dB.
g. Set the spectrum analyzer frequency control to 20.0 kHz and the frequency span/div. to 1 kHz. Observe and store the sweep on the spectrum analyzer.
h. Reset the input sensitivity of the spectrum analyzer to -10 dB. The system is now calibrated.
i. Disconnect the microphone with preamplifier from the input of the measuring amplifier.

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j. Disconnect the input to the spectrum analyzer from the measuring amplifier and connect the output from the transducer under test to the input of the spectrum analyzer. Connect a 6.8k ohm ±5 percent resistor across the transducer terminals.
k. Minimum acoustic bandwidth at the 3 dB points shall be 1.2 kHz.

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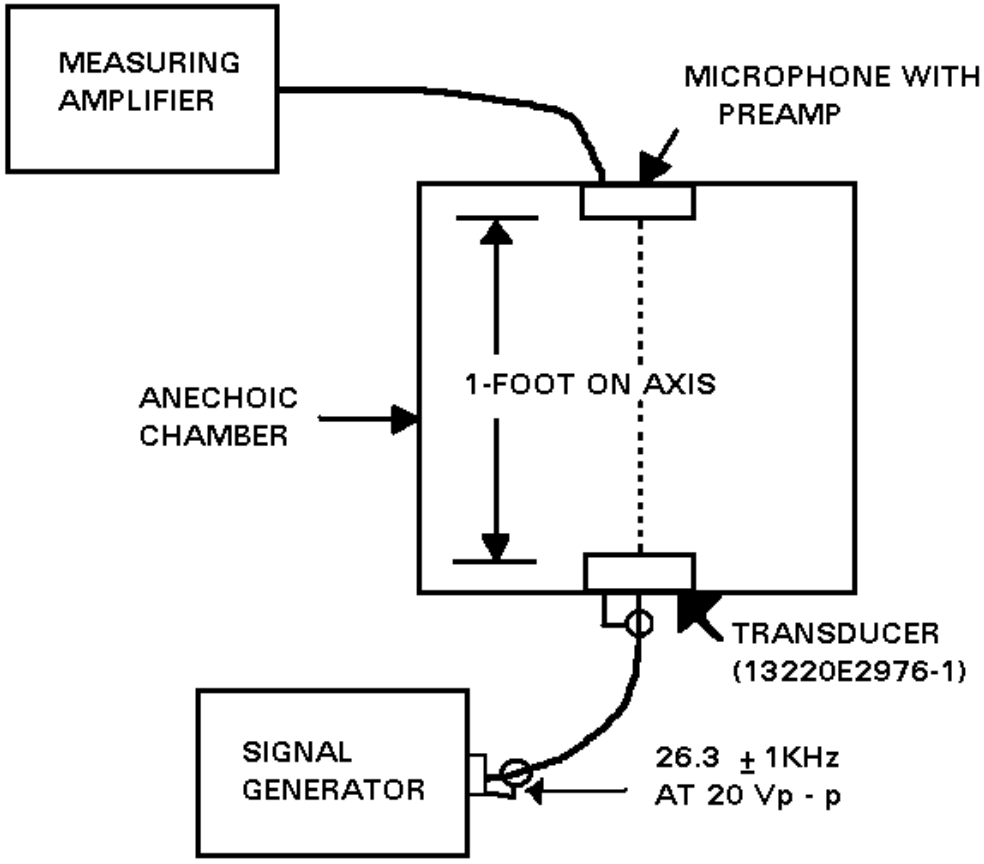


FIGURE 4. Functional test for transducer (13220E2976-1) white dot.

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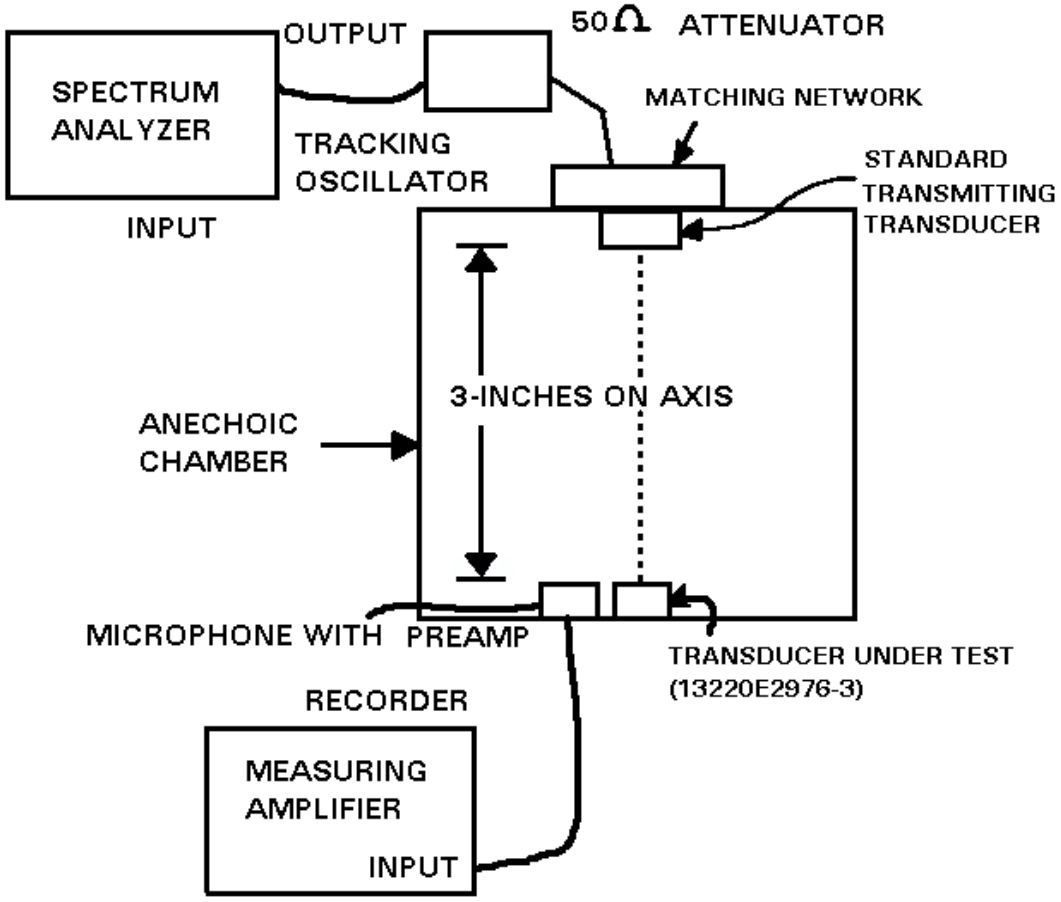


FIGURE 5. Functional test for transducer (13220E2976-3) blue dot.

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APPENDIX B

ENVIRONMENTAL TEST METHODS

B. SCOPE

B.1. Scope. The environmental test procedures contained in this appendix are to determine whether UMSs and major components furnished separately procured under this specification conform to the environmental requirements specified. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2. APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

B.3. GENERAL REQUIREMENTS

B.3.1. Test conditions. Unless otherwise specified herein, tests shall be performed under the following standard ambient conditions:

Temperature:	77 ±18 °F
Relative humidity:	50 ±30 percent.

B.3.1.1. Measurements of test conditions. All measurements of test conditions shall be made with instruments of the accuracy specified in B.3.1.3.

B.3.1.2. Tolerance of test conditions. Unless otherwise specified, tolerance of test conditions shall be as follows:

a. Air temperature at the control sensor: ±2.5 °F. The equipment sensor response time (T) shall be 20 seconds or less. Temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5 °F per foot in any direction, but never more than 4 °F total (equipment non-operating).

b. Relative humidity at the control sensor: ±5 percent.

NOTE: (T) is the time required for the sensing system to respond to 62.3 percent of a step change in temperature in the measured environment.

B.3.1.3. Accuracy of test apparatus. The accuracy of instruments and test equipment used to control or monitor the test parameters shall be verified and shall satisfy the requirements of ISO 10012-1 to the satisfaction of the procuring activity. All instruments and test equipment used in conducting the tests specified herein shall:

a. Conform to laboratory standards whose calibration is traceable to the prime standards at the US Bureau of Standards.

b. Have an accuracy of at least one-third the tolerance for the variable to be measured. In the event of conflict between this accuracy and a requirement for accuracy in any one of the test methods of this standard, the latter shall govern.

B.3.1.4. Stabilization of test temperature.

B.3.1.4.1. Operating. Unless otherwise specified, temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag is changing no more than 3.6 °F per hour.

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B.3.1.4.2. Non-operating. Unless otherwise specified, temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag reaches a temperature within 3.6 °F of the prescribed temperature. When changing temperatures, the temperature of the chamber air may be adjusted up to 9 °F beyond the desired end point for a period of time of up to 1 hour to reduce stabilization time, provided that the stabilization requirements of this paragraph are ultimately attained relative to the specified end point temperature, and provided the extended chamber temperatures will not cause damage to the test item.

B.3.2. Performance of test.

B.3.2.1. Functional test. The functional test referenced in the environmental test procedures shall be that specified in the applicable subparagraph of 4.4.2.5.

B.3.2.2. Pre-test functional test. Prior to proceeding with each environmental test, the test item shall be subjected to the applicable functional test (see B.3.2.1) under standard ambient conditions (see B.3.1). After the first environmental test (low temperature), this may be the post-test functional test (see B.3.2.5) after the previous environmental test.

B.3.2.3. Installation of test item in test chamber. The test chamber shall be at standard ambient conditions when the test item is installed. Suitable test leads shall be attached to each test item and brought out through port hole(s) to allow complete functional tests without opening the door of the test chamber during high temperature and low temperature tests. The test item shall then be subjected to the applicable functional test or an abbreviated functional test sufficient to determine that no malfunction or damage was caused due to faulty installation or handling.

B.3.2.4. Functional test at operating temperature. Perform applicable functional test (see B.3.2.1) during the test exposure as specified in the test procedure herein.

B.3.2.5. Post-test functional test. At the completion of each environmental test, the test item shall be subjected to the applicable functional test (see B.3.2.1) under standard ambient conditions (see B.3.1).

B.3.2.6. Failure criteria.

a. The item shall have failed the test when any of the following occur:

(1) Functional test results not in accordance with applicable test method during or after any of the tests.
(2) Deterioration, corrosion, damage or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational, service or maintenance requirements.

B.3.3. Test facilities and apparatus. Test facilities, chambers, and apparatus used in conducting the tests contained in this appendix shall be capable of meeting the conditions required.

B.3.3.1. Test chamber.

B.3.3.1.1. Volume of test chamber. The volume of the test chamber shall be such that the bulk of the item under test will not interfere with the generation and maintenance of the test conditions. When testing multiple

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sample items simultaneously, the test chamber shall be of sufficient size so that each test unit is provided uniform environmental conditions and is not subjected to non-test environments.

B.3.3.1.2. Heat source. The heat source of the test facility shall be so located that radiant heat from the source will not fall directly on the test item.

B.3.3.1.3. Location of temperature sensors. Unless otherwise specified, thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be located centrally within the chamber, in the supply air stream, or in the return air stream, whichever provides the specified test conditions at the bulk under test and shall be baffled or otherwise protected against radiation effects.

B.3.3.1.4. Internal air circulation. The conditioned air flow shall be suitably baffled to provide uniform airflow around the test item. If multiple test items are tested, they shall be so spaced as to provide free circulation between the test items and the chamber walls.

B.3.4. Test data. Test data shall include complete identification of all test equipment and accessories. The data shall include the actual test sequence used and ambient test conditions recorded periodically during the test period. The test record shall contain a signature and date block for certification of the test data by the test engineer.

B.3.5. Test sequence. Units being subjected to environmental tests shall have been functionally tested at room ambient temperature prior to environmental tests. Environmental tests shall be performed in the sequence low temperature, high temperature, humidity and shock, unless otherwise approved by the procuring contracting officer.

B.4. DETAILED TEST PROCEDURES

B.4.1. Low temperature test procedure.

B.4.1.1. Apparatus. Temperature chamber.

B.4.1.2. Procedure.

a. Prepare the test item(s) in accordance with general requirements, B.3.
b. Lower the internal chamber temperature to the storage temperature, -30 °F, and maintain for a period of 48 hours after stabilization.
c. Adjust the internal chamber temperature to -20 °F and maintain until temperature stabilization of the test item is reached (at least one hour after the chamber temperature sensor has reached -20 °F.
d. Apply operating power to the test item until the item is stabilized.
e. Perform the functional test of B.3.2.1.
f. Return the test item, non-operating, to standard ambient conditions and maintain for 2 hours or until temperature stabilization of the test item is reached.
g. Apply operating power, allow test item to stabilize, and perform the functional test conducted in step e. above (post-low temperature test).

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NOTE: The rate of temperature change during steps b., c., and f. may be the maximum attainable by the chamber but shall not exceed 18 °F per minute.

B.4.2. High temperature test procedure.

B.4.2.1. Apparatus. Temperature chamber.

B.4.2.2. Procedure.

a. Prepare the test item in accordance with general requirements, B.3.
b. Raise the internal chamber temperature to 120 °F.
c. Maintain internal chamber temperature for 6 hours at 120 °F.
d. Raise the internal chamber temperature to 165 °F within a time period of 1 hour and then maintain at that temperature for 4 additional hours.
e. Lower the internal chamber temperature to 120 °F within a time period of 1 hour.
f. Repeat steps c., d., and e. two additional times (making a total of three 12 hour cycles).
g. Adjust the internal chamber temperature to +150 °F and maintain until temperature stabilization of the test item is reached (minimum of one hour after chamber temperature has reached 150° F.
h. Apply power to the test item until the test item is stabilized and perform the functional test of Appendix A applicable to the test item.
i. Return the test item, non-operating, to standard ambient conditions and maintain for 2 hours or until temperature stabilization of the test item is reached.
j. Apply power to the test item until the item is stabilized and perform the test conducted in step h. above (post-high temperature test).

B.4.3. Humidity test procedure.

B.4.3.1. Apparatus. Humidity-temperature chamber and associated equipment.

B.4.3.1.1. Chamber. The chamber and accessories shall be constructed and arranged in such a manner as to avoid condensate dripping on the test item. The chamber shall be trap-vented to the atmosphere to prevent the buildup of total pressure. Relative humidity shall be determined from the dry bulb-wet bulb thermometer comparison method or an equivalent method approved by the procuring activity. When readout charts are used, they shall be capable of being read with a resolution within one degree F. When the wet bulb control method is used, the wet bulb and tank shall be cleaned and a new wick installed at least every 30 days. The air velocity flowing across the wet bulb shall be not less than 900 feet per minute. Provisions shall be made for controlling the flow of air throughout the internal chamber test space where the velocity of air shall not exceed 150 feet per minute. Steam or distilled, demineralized, or de-ionized water having a pH value between 6.0 and 7.2 at 73 °F shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the test item by the test facility.

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B.4.3.1.2. Procedure.

a. Prepare the test item in accordance with General Requirements, B.3. Prior to starting the test the internal chamber temperature shall be at standard ambient temperature (see B.3.1) with uncontrolled humidity.
b. Gradually raise internal chamber temperature to 149 °F and the relative humidity to 95 percent or greater over a period of 2 hours.
c. Maintain conditions of step b for not less than 6 hours.
d. Maintain 92 percent, or greater, relative humidity and reduce internal chamber temperature in 16 hours to 86 °F. Seventy five minutes before the end of the 16-hour period, commence performing a complete functional test of the test items in accordance with B.3.2.1. The test items may be removed from the environmental chamber for this test; however, unsuccessful completion of test or unsuccessful return of the test items to the test chamber within 75 minutes after initiation of tests shall constitute failure of the test. Failure to conform to the parameters specified in the functional test procedure shall also constitute failure of the test.
e. Repeat steps b., c., and d. for a total of 5 cycles (not less than 120 hours). Perform functional testing, as in step d above, commencing 75 minutes before the completion of each 24 hour cycle. Figure 6 is an outline of the humidity cycle for this procedure.
f. After the completion of the fifth test period, return test items to standard ambient conditions, allow test item to reach temperature stabilization and perform functional test as in step d.

NOTE: The equipment shall be operated only when required tests are being performed.

B.4.4. Shock test (bench handling). This procedure shall be used to determine the ability of equipment to withstand the shock encountered during servicing. Remove the cover to perform test. The test item shall be placed in a suitable position for servicing on a horizontal, solid wooden bench top at least 1-5/8 inches thick. The test shall be performed, as follows, in a manner simulating shocks likely to occur during servicing.

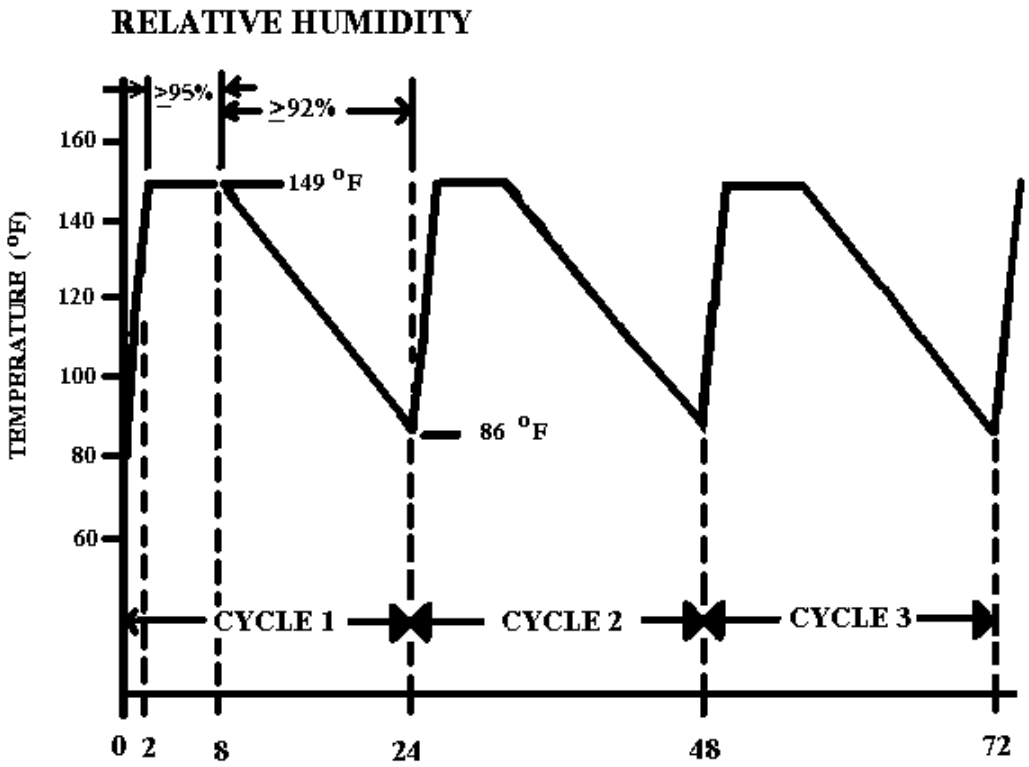
Step 1. Using one edge as a pivot, lift the opposite edge of the chassis until one of the following conditions occurs (whichever occurs first):

a. The chassis forms an angle of 45 degrees with the horizontal bench top.
b. The lifted edge of the chassis has been raised 4 inches above the horizontal bench top.
c. The lifted edge of the chassis is just below the point of perfect balance.

Let the chassis drop back freely to the horizontal bench top. Repeat, using other practical edges of the same horizontal face as pivot points, for a total of four drops.

Step 2. Repeat step 1, with the test item resting on other faces until it has been dropped for a total of four times on each face on which the test item could be placed practicably during servicing. The test item shall not be operating during the test. At the conclusion of the test, the test item shall be inspected and subjected to the functional test of B.3.2.1. Failure criteria shall be as specified in B.3.2.5.

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NOTE: CONTINUE FOR A TOTAL OF 5 CYCLES (120 HOURS)

FIGURE 6. Test cycle for humidity test.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-52872D	**2. DOCUMENT DATE (YYMMDD)** 98/07/17

3. DOCUMENT TITLE SENIOR GROUP, ULTRASONIC MOTION: TRANSCEIVER RT-1161/FSS-(V) AND PROCESSOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

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